

MEDIATEK

MT8676 Android Camera Turbo User Manual

Version: 1.1
Release date: 2024-11-12

Use of this document and any information contained therein is subject to the terms and conditions set forth in Exhibit 1. This document is subject to change without notice.

Version History

Version	Date	Author	Description
1.0	2024-08-12	Houchu.Fan	Official release
1.1	2024-11-12	Houchu.Fan	Added Section 1.3 Camera Common Questions/Problem Analysis

Table of Contents

Version History 2

Table of Contents 3

List of Figures 3

1 Camera Turbo 4

1.1 Camera Architecture Overview 4

1.2 Camera Turbo 5

1.2.1 Architecture 5

1.2.2 Control Flow 6

1.3 Camera Common Questions/Problem Analysis 7

1.3.1 Camera Debug Log 7

1.3.2 Camera Buffer Dump 8

1.3.2.1 Raw Sensor 8

1.3.2.2 YUV Sensor 8

1.3.3 Camera Sensor Info 8

1.3.4 Camera Basic Flow 9

1.3.4.1 Open 9

1.3.4.2 Configure 10

1.3.4.3 Flush 10

1.3.4.4 Close 10

1.3.4.5 Sensor Power-on Done 10

Exhibit 1 Terms and Conditions 11

List of Figures

Figure 1-1. Camera architecture overview 4

Figure 1-2. Camera MW architecture 5

Figure 1-3. MW class flow 6

Figure 1-4. MW class flow 7

1 Camera Turbo

1.1 Camera Architecture Overview

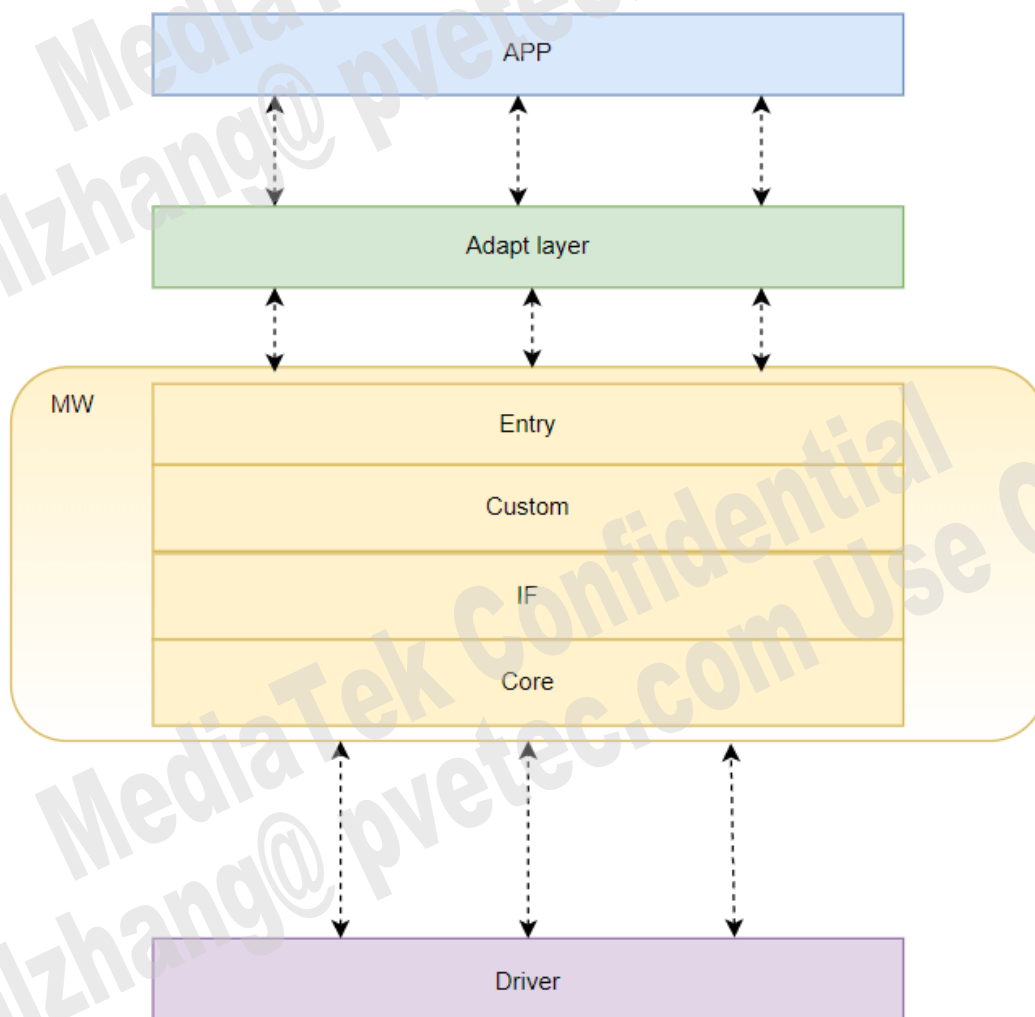


Figure 1-1. Camera architecture overview

Camera Turbo is mainly divided into four parts:

- Entry layer is the entrance to MW. Different OSs can adapt MW by calling the code of the entry layer through the adapt layer.
- Custom is the customization layer through which customers can perform customized operations.
- The IF layer is the interface layer used by the entry layer. The logic of camera session and native camera is generated here. The IF layer mainly implements the flow framework of request and config-related operations.
- The core layer is the core layer of MW. It mainly communicates with the driver, sends the request and gets the result back, and finally calls back to the APP through the IF layer.

1.2 Camera Turbo

1.2.1 Architecture

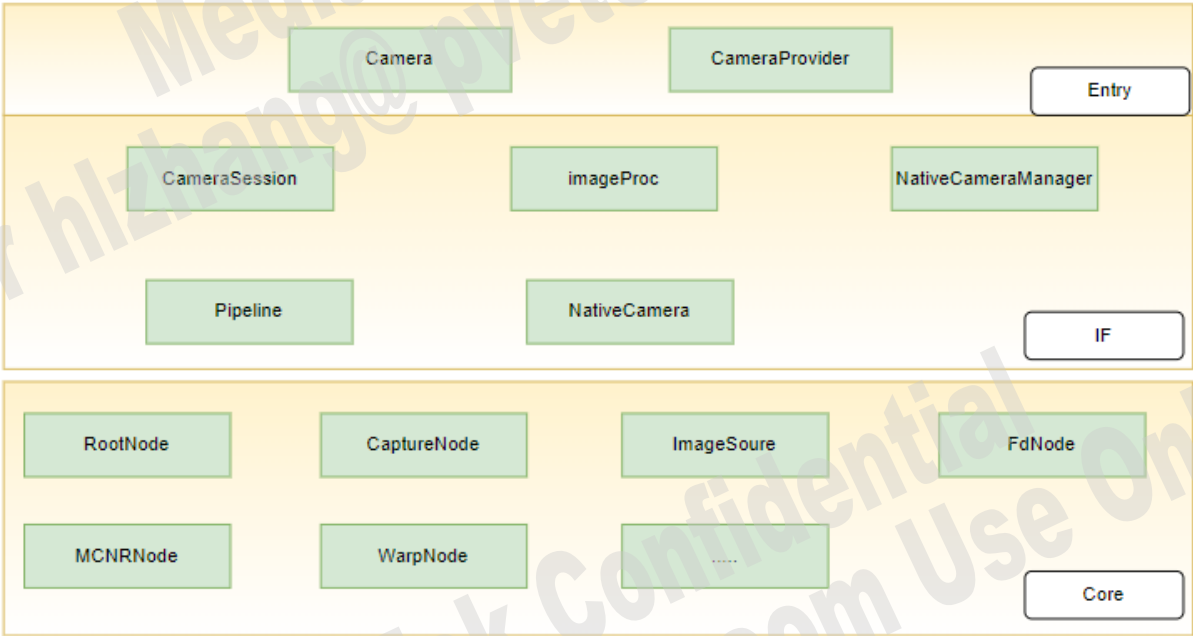


Figure 1-2. Camera MW architecture

The main architecture diagram of Camera Turbo is the entry layer, IF layer and Core layer. The custom layer is mainly for customer use. If there is no need to use it, bypass it directly.

- The entry layer is mainly composed of Camera and CameraProvider. The CameraProvider mainly provides the Camera list to the upper layer and obtains and controls the properties of the Camera. The Camera mainly provides the interface to the upper layer to control the control behavior of the camera sensor.
- The IF layer is mainly the pipeline, which is the manager that builds the underlying node connection relationship. ImageProc mainly facilitates callback to provide customers with customized behaviors. At the same time, customers can also customize the pipeline through ImageProc.
- The core layer is mainly composed of subclasses of ImageNode. The subclasses interact with the underlying driver to implement different operations on the request. For example, FdNode is mainly for face recognition, captureNode for camera function, MCNRNode for image processing, and the final result will be passed through ImageNode callback to IF layer.

1.2.2 Control Flow

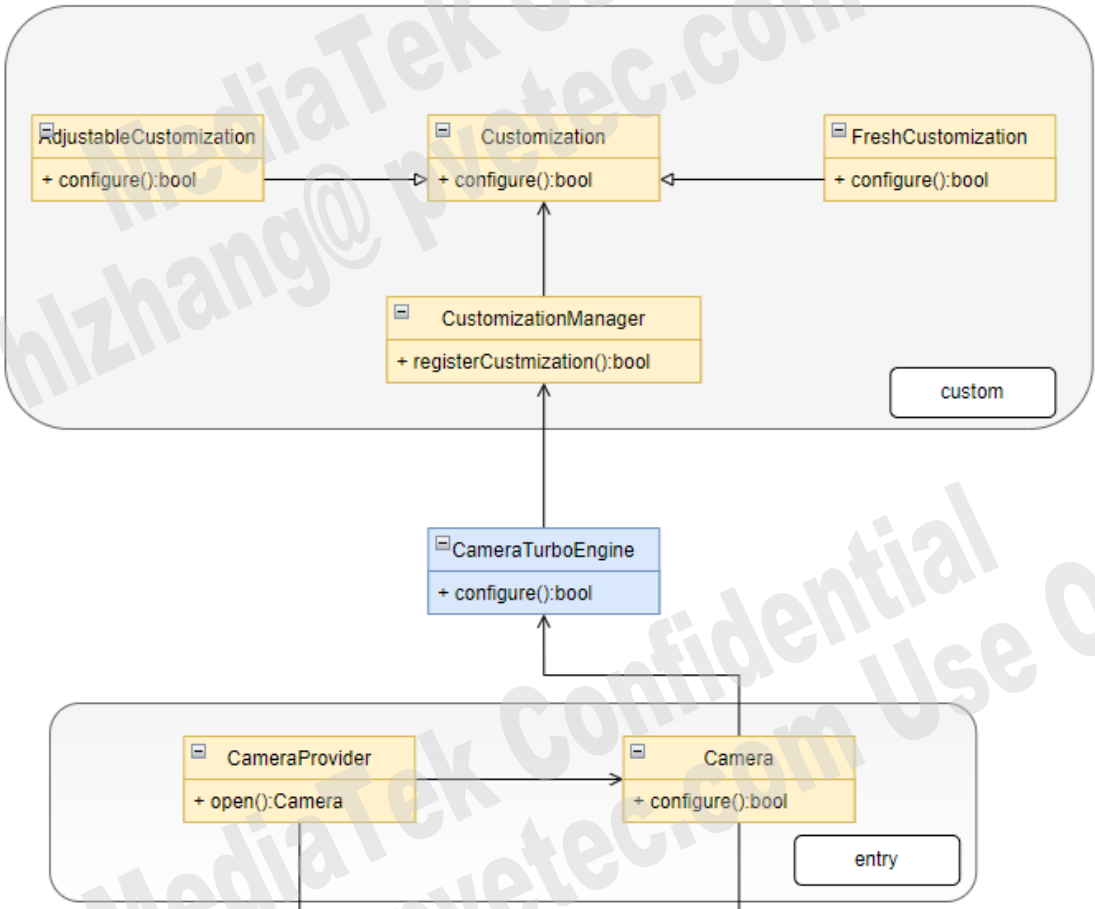


Figure 1-3. MW class flow

Figure 1-3 is mainly the class flow interaction diagram between entry and custom. It mainly tells that the upper layer gets the Camera through the open interface of CameraProvider, and the Camera gets the CustomizationManger by calling Camera TurboEngine. CustomizationManger is the manager of the custom layer, which facilitates the management of customer-specific customization behaviors.

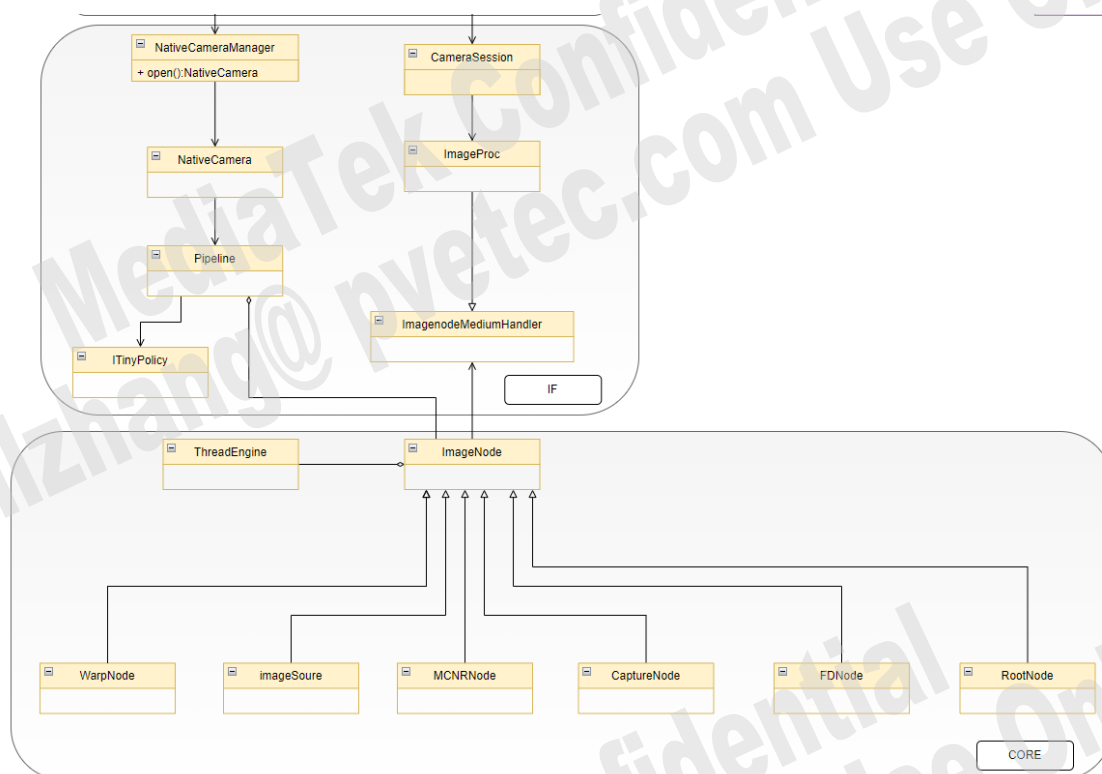


Figure 1-4. MW class flow

The interaction logic between the IF layer and CORE is mainly that the open interface of the upper CameraProvider calls the NativeCameraManager. The NativeCameraManager allocates the NativeCamera that can be operated at the bottom layer. Each NativeCamera has pipeline attributes, which determines the actual flows of each NativeCamera. The Camera's configuration will create a new CameraSession. CameraSession creates its own Pipeline. The connected ImageProc is saved in the Pipeline. ImageProc will create the corresponding ImageNode. Each CameraSession has its own ImageProc list. Users can perform customized behaviors in ImageProc. ImageProc will use MediumHandler, which is the callback interface. ImageNode callback will use ImageProc.

1.3 Camera Common Questions/Problem Analysis

1.3.1 Camera Debug Log

```
adb shell setprop persist.mtk.camera.log_level N
```

N can set be to 0 ~ 4, default value is 3.

```
0/1 : LOGE/LOGW LEVEL
2 : LOGI LEVEL
3 : LOG LEVEL
4 : LOGV LEVEL
```

Note: Set this CMD, and you need to reboot camerahalservice.

1.3.2 Camera Buffer Dump

1.3.2.1 Raw Sensor

```
// before open camera
adb root
mkdir -p /data/vendor/camera_dump
adb shell setprop vendor.debug.camera.vrp.dump 2
// running (1:start dump; 0: stop dump)
adb shell setprop vendor.debug.camera.vrp.dump.start N
```

1.3.2.2 YUV Sensor

```
// before open camera
adb root
mkdir -p /data/vendor/camera_dump
chmod 777 /data/vendor/camera_dump
// running (2 is start dump, 0 is stop dump)
adb shell setprop vendor.debug.camera.coredevice.wpe.dump 2

it dump isp p2(pq dip) iuput/output(isp p2 input is isp p1(camsv) output)
```

```
-rwx----- 1 cameraserver audio 19660800 2024-11-11 17:08 wpepq98483-0000-in-IMGI-PW7680x1280-15360.UYVY
-rwx----- 1 cameraserver audio 777600 2024-11-11 17:08 wpepq98483-0000-out-WDMAO-PW960x540-960.NV21
-rwx----- 1 cameraserver audio 19660800 2024-11-11 17:08 wpepq98483-0001-in-IMGI-PW7680x1280-15360.UYVY
-rwx----- 1 cameraserver audio 777600 2024-11-11 17:08 wpepq98483-0001-out-WDMAO-PW960x540-960.NV21
-rwx----- 1 cameraserver audio 19660800 2024-11-11 17:08 wpepq98483-0002-in-IMGI-PW7680x1280-15360.UYVY
-rwx----- 1 cameraserver audio 777600 2024-11-11 17:08 wpepq98483-0002-out-WDMAO-PW960x540-960.NV21
-rwx----- 1 cameraserver audio 19660800 2024-11-11 17:08 wpepq98483-0003-in-IMGI-PW7680x1280-15360.UYVY
-rwx----- 1 cameraserver audio 777600 2024-11-11 17:08 wpepq98483-0003-out-WDMAO-PW960x540-960.NV21
```

1.3.3 Camera Sensor Info

Dump sensor info CMD is as follows:

```
adb shell sentest_v412
```

```

[main]sizeof long : 4
[debug][init_sensor_list] ioctl fail, line=300
[show Sensors]sensorNum 7
[show Sensors]name:SENSOR_DRVNAME_0X08B40_MIPI_RAW type:0 max_scen: 8
[show Sensors]index:0, SensorDevIdx:1
[show Sensors]master_dts_idx:0, vc_num:0
[show Sensors]name:SENSOR_DRVNAME_MAX96712_MIPI_YUV type:1 max_scen: 4
[show Sensors]index:1, SensorDevIdx:2
[show Sensors]master_dts_idx:0, vc_num:0
[show Sensors]name:SENSOR_DRVNAME_MAX96712A_MIPI_YUV type:1 max_scen: 4
[show Sensors]index:2, SensorDevIdx:16
[show Sensors]master_dts_idx:4, vc_num:4
[show Sensors]slave_dts_idx:6, vc_id:0, link_status:0
[show Sensors]slave_dts_idx:7, vc_id:1, link_status:0
[show Sensors]slave_dts_idx:8, vc_id:2, link_status:0
[show Sensors]slave_dts_idx:9, vc_id:3, link_status:0
[show Sensors]name:SENSOR_DRVNAME_MAX96712A0_MIPI_YUV type:1 max_scen: 4
[show Sensors]index:3, SensorDevIdx:64
[show Sensors]master_dts_idx:4, vc_num:4
[show Sensors]slave_dts_idx:6, vc_id:0, link_status:0
[show Sensors]slave_dts_idx:7, vc_id:1, link_status:0
[show Sensors]slave_dts_idx:8, vc_id:2, link_status:0
[show Sensors]slave_dts_idx:9, vc_id:3, link_status:0
[show Sensors]name:SENSOR_DRVNAME_MAX96712A1_MIPI_YUV type:1 max_scen: 4
[show Sensors]index:4, SensorDevIdx:128
[show Sensors]master_dts_idx:4, vc_num:4
[show Sensors]slave_dts_idx:6, vc_id:0, link_status:0
[show Sensors]slave_dts_idx:7, vc_id:1, link_status:0
[show Sensors]slave_dts_idx:8, vc_id:2, link_status:0
[show Sensors]slave_dts_idx:9, vc_id:3, link_status:0
[show Sensors]name:SENSOR_DRVNAME_MAX96712A2_MIPI_YUV type:1 max_scen: 4
[show Sensors]index:5, SensorDevIdx:256
[show Sensors]master_dts_idx:4, vc_num:4
[show Sensors]slave_dts_idx:6, vc_id:0, link_status:0
[show Sensors]slave_dts_idx:7, vc_id:1, link_status:0
[show Sensors]slave_dts_idx:8, vc_id:2, link_status:0
[show Sensors]slave_dts_idx:9, vc_id:3, link_status:0
[show Sensors]name:SENSOR_DRVNAME_MAX96712A3_MIPI_YUV type:1 max_scen: 4
[show Sensors]index:6, SensorDevIdx:512
[show Sensors]master_dts_idx:4, vc_num:4
[show Sensors]slave_dts_idx:6, vc_id:0, link_status:0
[show Sensors]slave_dts_idx:7, vc_id:1, link_status:0
[show Sensors]slave_dts_idx:8, vc_id:2, link_status:0
[show Sensors]slave_dts_idx:9, vc_id:3, link_status:0
[main]Param: 1 <sensorDev> <scenario> <fps>
[main]<sensorDev> : testmdl(0), main(1), Sub(2), Main2(4), sub2(8), Main3(16)
[main]<scenario> : Pre(0), Cap(1), VD(2), slim1(3), slim2(4), or TM pattern id
[main]<Test ID>: (total test case 0)
[main]ID: 000 Test_stream on
[main]ID: 001 Test_mipi_rate
[main]ID: 002 Test_raw_dump
[main]ID: 003 Test_fps
[main]ID: 005 Test_gain_linearity
[main]ID: 006 Test_normal_shutter_linearity
[main]ID: 007 Test_flicker
[main]ID: 009 Test_perframe_control
[main]ID: 010 Test_long_exposure_linearity
[main]ID: 011 Test_i2c_performance
[main]ID: 012 Test_mode_stability
[main]ID: 013 Test_sensor_drv_monkey
[main]ID: 050 Test_manual_gain_range_linearity
[main]ID: 051 Test_manual_shutter_range_linearity
[main]ID: 060 SensorInfo Dump
[main]ID: 061 Manual raw dump
[main]ID: 100 Test_min_shutter_linearity
[main]ID: 101 Test_Gain_table_linearity
[main]ID: 102 Test_binning_ratio

```

When camera cannot be found, you should input this CMD first to check sensor info.

1.3.4 Camera Basic Flow

1.3.4.1 Open

```

10-20 09:20:26.102 25166 25166 D TINY_CAMERA PROVIDER: [open] tinylib version:V1.0
10-20 09:20:26.102 25166 25166 D TINY_CAMERA PROVIDER: [open] open virtual 1, real 1
10-20 09:20:26.104 25166 25166 D TINY_CAMERA PROVIDER: [open] open camera Id:1 realId:1 userId:0 0xb4000075af54f098

```

1.3.4.2 Configure

```

10-20 09:20:26.232 25166 25220 D TINY_CAMERA: [1][0][configure] +
10-20 09:20:26.232 25166 25220 D TINY_CAMERA: [1][0][configure] stream id 0, size (960x540), fmt (11)
10-20 09:20:26.232 25166 25220 D TINY_CAMERA: [1][0][configure] streamId:[ap:0 -> hal:0]
10-20 09:20:26.232 25166 25220 D TINY_CAMERA: [1][0][configure] stream id 1, size (1280x720), fmt (21)
10-20 09:20:26.232 25166 25220 D TINY_CAMERA: [1][0][configure] streamId:[ap:1 -> hal:1]
10-20 09:20:26.232 25166 25220 D TINY_CAMERA: [1][0][configure] stream id 2, size (1280x720), fmt (32315659)
10-20 09:20:26.232 25166 25220 D TINY_CAMERA: [1][0][configure] streamId:[ap:2 -> hal:2]
10-20 09:20:26.232 25166 25220 D TINY_CAMERA: [1][0][configure] create sessionId:0
10-20 09:20:26.232 25166 25220 D TINY_CAMERA: [configure] create sessionId:0
10-20 09:20:26.481 25166 25220 D TINY_CAMERA: [1][0][configure] [out] streamId[hal:0 -> ap:0]
10-20 09:20:26.481 25166 25220 D TINY_CAMERA: [1][0][configure] [out] streamId[hal:1 -> ap:1]
10-20 09:20:26.481 25166 25220 D TINY_CAMERA: [1][0][configure] [out] streamId[hal:2 -> ap:2]
10-20 09:20:26.481 25166 25220 D TINY_CAMERA: [1][0][configure] [in] streamId[hal:0 -> ap:0]
10-20 09:20:26.481 25166 25220 D TINY_CAMERA: [1][0][configure] [in] streamId[hal:1 -> ap:1]
10-20 09:20:26.481 25166 25220 D TINY_CAMERA: [1][0][configure] [in] streamId[hal:2 -> ap:2]
10-20 09:20:26.481 25166 25220 D TINY_CAMERA: [1][0][configure] -

```

1.3.4.3 Flush

```

10-20 09:20:32.121 25166 25166 D TINY_CAMERA: [1][0][flush] +
10-20 09:20:32.442 25166 25166 D TINY_CAMERA: [1][0][flush] -

```

1.3.4.4 Close

```

10-20 09:20:32.455 25166 25166 D TINY_CAMERA: [1][0][close] +
10-20 09:20:32.455 25166 25166 D TINY_CAMERA: [1][0][close] -
10-20 09:20:32.534 25166 25166 D TINY_CAMERA: [1][0][~Camera] dtor 0xb4000075af54f098

```

1.3.4.5 Sensor Power-on Done

```

10-20 09:20:26.256 25166 25220 D TINY_NATIVECAMERA: [1][waitUntilPowerOnDone] waitUntilPowerOnDone +
10-20 09:20:26.455 25166 25220 D TINY_NATIVECAMERA: [1][waitUntilPowerOnDone] waitUntilPowerOnDone -

```

Exhibit 1 Terms and Conditions

Your access to and use of this document and the information contained herein (collectively this “Document”) is subject to your (including the corporation or other legal entity you represent, collectively “You”) acceptance of the terms and conditions set forth below (“T&C”). By using, accessing or downloading this Document, You are accepting the T&C and agree to be bound by the T&C. If You don’t agree to the T&C, You may not use this Document and shall immediately destroy any copy thereof.

This Document contains information that is confidential and proprietary to MediaTek Inc. and/or its affiliates (collectively “MediaTek”) or its licensors and is provided solely for Your internal use with MediaTek’s chipset(s) described in this Document and shall not be used for any other purposes (including but not limited to identifying or providing evidence to support any potential patent infringement claim against MediaTek or any of MediaTek’s suppliers and/or direct or indirect customers). Unauthorized use or disclosure of the information contained herein is prohibited. You agree to indemnify MediaTek for any loss or damages suffered by MediaTek for Your unauthorized use or disclosure of this Document, in whole or in part.

MediaTek and its licensors retain titles and all ownership rights in and to this Document and no license (express or implied, by estoppels or otherwise) to any intellectual propriety rights is granted hereunder. This Document is subject to change without further notification. MediaTek does not assume any responsibility arising out of or in connection with any use of, or reliance on, this Document, and specifically disclaims any and all liability, including, without limitation, consequential or incidental damages.

THIS DOCUMENT AND ANY OTHER MATERIALS OR TECHNICAL SUPPORT PROVIDED BY MEDIATEK IN CONNECTION WITH THIS DOCUMENT, IF ANY, ARE PROVIDED “AS IS” WITHOUT WARRANTY OF ANY KIND, WHETHER EXPRESS, IMPLIED, STATUTORY, OR OTHERWISE. MEDIATEK SPECIFICALLY DISCLAIMS ALL WARRANTIES OF MERCHANTABILITY, NON-INFRINGEMENT, FITNESS FOR A PARTICULAR PURPOSE, COMPLETENESS OR ACCURACY AND ALL WARRANTIES ARISING OUT OF TRADE USAGE OR OUT OF A COURSE OF DEALING OR COURSE OF PERFORMANCE. MEDIATEK SHALL NOT BE RESPONSIBLE FOR ANY MEDIATEK DELIVERABLES MADE TO MEET YOUR SPECIFICATIONS OR TO CONFORM TO A PARTICULAR STANDARD OR OPEN FORUM.

Without limiting the generality of the foregoing, MediaTek makes no warranty, representation or guarantee regarding the suitability of its products for any particular purpose, nor does MediaTek assume any liability arising out of the application or use of any product, circuit or software. You agree that You are solely responsible for the designing, validating and testing Your product incorporating MediaTek’s product and ensure such product meets applicable standards and any safety, security or other requirements.

The above T&C and all acts in connection with the T&C or this Document shall be governed, construed and interpreted in accordance with the laws of Taiwan, without giving effect to the principles of conflicts of law.